

DC014

Computer Science 1

Semester 1

2023/2024

2 hours

DC014

Sains Komputer 1

Semester 1

Sesi 2023/2024

2 jam

Nama :												Tutoran :											
Nama Pensyarah Tutoran :																							
No. Matrik												No. Kad Pengenalan											
M	S																						



KOLEJ MATRIKULASI PERAK
PERAK MATRICULATION COLLEGE

PRA PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME TRIAL EXAMINATION

SAINS KOMPUTER
2 jam

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Untuk Kegunaan Pemeriksa		
No. Soalan	Markah	Markah Pemeriksa
1		
2		
3		
4		
5		
6		
JUMLAH		

INSTRUCTIONS TO CANDIDATE:

This question paper consists of **6** questions.

Answer **all** questions in the space provided.

The use of non-programmable scientific calculator is permitted.

ARAHAN KEPADA CALON:

*Kertas soalan ini mengandungi **6** soalan.*

*Jawab **semua** soalan di ruang yang disediakan.*

Penggunaan kalkulator saintifik yang tidak boleh diprogramkan dibenarkan

1. You are creating a multimedia presentation for your school science project. You want to include a variety of multimedia elements to make your presentation more engaging and informative.

- (a) Describe **one (1)** multimedia element, which you should include in your multimedia presentation.

[2 marks]

- (b) Explain the purpose of the multimedia element mentioned in 1(a).

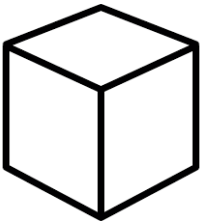
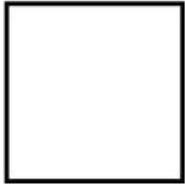
[2 marks]

- (c) You also use bitmap images as elements in your presentation. Describe the advantage of using bitmap.

[2 marks]

2. (a) (i) Based on the shape in the table below, compare specifically the difference between dimensional in multimedia application representation in terms of axis and dimension.

[2 marks]

Shape		
Axis		
Dimension		

- (ii) This type of multimedia application representation uses 3D film with added effects that take place in specially designed theaters.

Based on the above statement, explain about the file size of the multimedia application representation.

[2 marks]

- (b) A good multimedia project is built through 3 phases of development namely: Pre-Production, Production and Post-Production.

Study the situation below and state the phases of multimedia design and the activities involved

[6 marks]

Situation	Phase and activity
(i) Raju used MS PowerPoint software to combine all the multimedia elements to complete his presentation assignment	Phase: Activity:
(ii) Tests that have been carried out on several computer units have shown that the multimedia product needs to be improved	Phase: Activity:
(iii) These rough sketches bring together all the elements with the controls and depict the sequence of the action.	Phase: Activity:

- (c) Explain **two (2)** types of authoring tools that exist in **Figure 1**

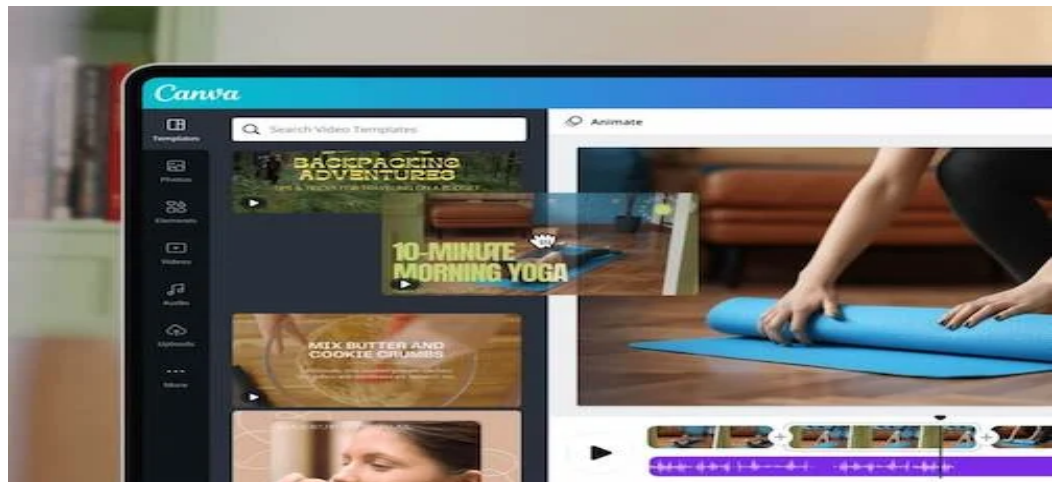


Figure 1

[4 marks]

Type of Authoring Tool	Characteristic

- (d) YouTube has 2.1 billion monthly active users around the world. That number shows no signs of slowing down, with the projected amount of users increasing each year.

Describe **one (1)** features based on content and accessibility to describe YouTube as a medium of delivery

[2 marks]

Content	Accessibility

3. (a)

- Each department manages their own database
- Database is design for single or limited number of users
- Data collected are usually small in size/ number

State the most suitable approach in managing and storing data, base on the statements above.

[1 mark]

(b) Describe **one (1)** weakness in using the approach stated above.

[2 marks]

(c) (i) Name another approach in managing and storing data other than the one stated in 3(a).

[1 mark]

(ii) List **two (2)** disadvantage of using database system.

[2 marks]

4. (a)

Data integrity is an important component of data quality. You can't have high quality data if you don't have data integrity.

The preceding statement underscores the significance of data quality and integrity.

Give **two (2)** characteristics used to assess data integrity.

[2 marks]

- (b) Determine which database object is best suited for use in the following scenario. [2 marks]

Scenario	Suitable database object
Encik Ahmad, a passenger, wants to enter his information into the airline database before confirming his reservations.	
En Ahmad, after finishing filling in his particulars in the database, proceed to make a hardcopy of his booking	

5. Network and the internet are very close to our society today. Not only from phone calls, network use at home and business networks, network and internet are very essential.

Briefly explain the uses of networks in the following areas.

[4 marks]

Areas	Explanation
(a) Telemedicine	
(b) Global Positioning System (GPS)	

- (c) State the type of wired transmission media based on given usage.

[4 marks]

Usage	Type of wired transmission media
Commonly used for a network such as a computer laboratory.	
Can support higher data rates and longer distances and is commonly used in cable television networks.	
Offers the highest data rates and longest distances and commonly used long-distance communication	
Relatively inexpensive and easy to install, which makes it ideal for a variety of applications such as in telephone networks.	

- (d) State the type of wireless transmission media based on the given characteristics. [4 marks]

Characteristics	Type of wireless transmission media
The longest wavelength on the electromagnetic spectrum, and they can travel long distances through air and penetrate obstacles.	
They are commonly used for satellite communications and cellular telephone networks.	
It is commonly used for short-range communication between devices such as remote controls.	
Bluetooth is a short-range wireless technology that connects devices such as smartphones, headphones, and speakers.	

- (e) Identify the communication device based on the given function. [6 marks]

Function	Communication device
Connects devices on the same network and sends all data packets to all connected devices.	
Converts digital signals from a computer into analog signals that can be transmitted over a telephone line or other communication channel.	
Allows devices to connect to a network wirelessly.	
Allows a device to connect to a network.	
Extends the range of a network signal.	
Connects different networks together and forwards data packets to their destination.	

6. (a)

- Using this search tool, user can locate a particular topic by clicking links through different levels, moving from the general to the specific.
- users have difficulty deciding which categories to choose as they work through the menus of links presented

Identify the type of search tool mentioned above and justify your answer.

[2 marks]

(b) A search operator is a character or string of characters used in a search engine query to narrow the focus of the search.

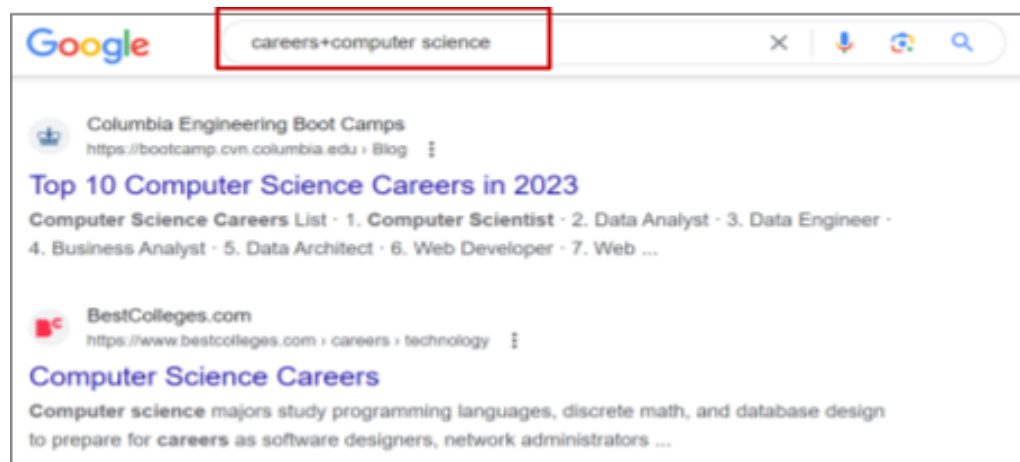


Figure 3

(i) Identify the search engine operator used in the **Figure 3** given above. Justify your answer.

[2 marks]

(ii) Name another search operator other than the answer given in b (i) and justify your answer.

[2 marks]

A collage of various social media icons including Twitter, Facebook, Instagram, YouTube, LinkedIn, and others.

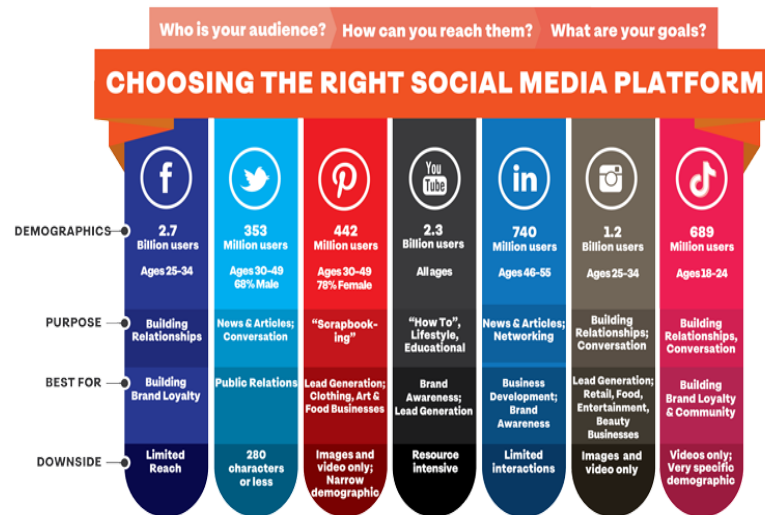
Figure 4

Social media in **Figure 4** is a website that encourages members in its online community to share their interests, ideas, stories, photos, music and videos with others registered users.

- (i) Describe purposes based on table below.

[2 marks]

Examples of Social Media	Descriptions
	
	



(ii)

Figure 5

Mahmoud et al. (2020) identified various capabilities (including relational and informational) that facilitate the successful deployment and transformation of social media marketing-related resources.

<https://onlinelibrary.wiley.com/doi/full/10.1002/mar.21746>

Identify and explain type of web site activity that can be linked to social media to buying and selling of goods and services, or the transmitting of funds or data, over an electronic network, primarily the internet.

[2 marks]

- (d) (i) Censorship is the suppression of speech, public communication, or other information. This may be done on the basis that such material is considered objectionable, harmful, sensitive, or "inconvenient".

Based on your understanding, describe one purpose of censorship.

[2 marks]

- (ii) Using your own words, explain the affects of Internet censorship on individuals.

[2 marks]

- (e) Data privacy is the relationship between the collection and dissemination of data, technology, the public expectation of privacy, and the legal and political issues surrounding them.

Explain two privacy issues concerning the Internet.

[4 marks]

- (f) The Internet of Things (IoT), is a network of interrelated devices that connect and exchange data with other IoT devices and the cloud. IoT devices are typically embedded with technology such as sensors and software and can include mechanical and digital machines and consumer objects.

Describe one benefit of IoT.

[2 marks]

END OF QUESTION PAPER